MTE 204 Problem Set #5: Solutions

Integration and Differentiation

(Chapter 21, 22 and 23 in Chapra 8th edition, International Version)

21.6 Integrate the following function both analytically and numerically. Use both the trapezoidal and Simpson's 1⁄3 rules to numerically integrate the function. For both cases, use the multiple-application version, with $n = 4$. Compute percent relative errors for the numerical results.

$$\int_0^3 x^2 e^x \, dx$$

Analytical solution:

$$\int_0^3 x^2 e^x \, dx = [(x^2 - 2x + 2)e^x]\,_0^3 = 98.42768$$

Trapezoidal rule (n = 4):

Recall:

$$I = \underbrace{(b-a)}_{\text{width}} \underbrace{\frac{f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)}{2n}}_{\text{Average height}}$$

For n=4, step size =0.75

$$I = (3-0)\frac{0 + 2(1.190813 + 10.0838 + 48.03166) + 180.7698}{8} = 112.2684 \qquad \varepsilon_t = 14.062\%$$

Alternatively:

$$\int_0^3 x^2 e^x \, dx$$

Trapezoid rule, n=4		
x	f(x)	Area calculation
0	0	
0.75	1.191	0.447
1.50	10.084	4.228
2.25	48.032	21.793
3.00	180.770	85.801
	total area ->	112.268

Simpson's rule (n = 4):

$$I = (3-0)\frac{0 + 4(1.190813 + 48.03166) + 2(10.0838) + 180.7698}{12} = 99.45683 \qquad \varepsilon_t = 1.046\%$$

21.9 Suppose that the upward force of air resistance on a falling object is proportional to the square of the velocity. For this case, the velocity can be computed as

$$v(t) = \sqrt{\frac{gm}{c_d}}\ tanh\left(\sqrt{\frac{gc_d}{m}}t\right)$$

where c_d = a second-order drag coefficient. **(a)** If g = 9.81 m/s^2, m = 68.1 kg, and c_d = 0.25 kg/m, use analytical integration to determine how far the object falls in 10 s. **(b)** Make the same evaluation, but evaluate the integral with the multiple-segment trapezoidal rule. Use a sufficiently high n that you get three significant digits of accuracy.

Analytical solution:

$$z(t) = \int_0^t \sqrt{\frac{gm}{c_d}} tanh\left(\sqrt{\frac{gc_d}{m}}t\right) dt = \left[\frac{m}{c_d} ln\left[cosh\left(\sqrt{\frac{gc_d}{m}}t\right)\right]\right]_0^t$$

$$z(10) = \left[\frac{68.1}{0.25} ln\left[cosh\left(\sqrt{\frac{9.81(0.25)}{68.1}}(10)\right)\right]\right]_0^{10} = 334.1782$$

Thus, the result to 3 significant digits is 334. Here are results for various multiple-segment trapezoidal rules:

n	*I*
1	247.1068
2	314.6304
3	325.7253
4	329.4623
5	331.1708
6	332.0937
7	332.6485
8	333.0079
9	333.254
10	333.4298
11	333.5599

Thus, an 11-segment application gives the result to 3 significant digits.

21.11 Evaluate the integral of the following tabular data with **(a)** the trapezoidal rule and **(b)** Simpson's rules:

x	−2	0	2	4	6	8	10
$f(x)$	35	5	−10	2	5	3	20

(a) Trapezoidal rule (n = 6):

$$I = (10-(-2))\frac{35+2(5-10+2+5+3)+20}{12} = 65$$

(b) Simpson's rules (n = 6):

$$I = (10-(-2))\frac{35+4(5+2+3)+2(-10+5)+20}{18} = 56.66667$$

22.2 Use Romberg integration to evaluate

$$I = \int_1^2 \left(x+\frac{1}{x}\right)^2 dx$$

to an accuracy of ε_s = 0.5% based on Eq. (22.9). Your results should be presented in the form of Fig. 22.3. Use the analytical solution of the integral to determine the percent relative error of the result obtained with Romberg integration. Check that ε_t is less than the stopping criterion ε_s.

The integral can be evaluated analytically as,

$$I = \int_1^2 \left(x+\frac{1}{x}\right)^2 dx = \int_1^2 x^2+2+x^{-2}\ dx$$

$$I = \left[\frac{x^3}{3}+2x-\frac{1}{x}\right]_1^2 = \frac{2^3}{3}+2(2)-\frac{1}{2}-\frac{1^3}{3}-2(1)+\frac{1}{1} = 4.8333$$

The tableau depicting the implementation of Romberg integration to ε_a = 0.5% is

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Question 22.2

$$I = \int_1^2 \left(x + \frac{1}{x}\right)^2 dx$$

Exact answer-> 4.8333

ε_t

iteration ->		1	2	3
error (true) %		6.035	0.096	0.004
error (appox) %			5.933	0.093

ε_a

# segments	h	Integral (Trapezoid)	Romberg	Romberg
1	1	5.1250	4.83796296	4.83347
2	0.5	4.9097	4.83375094	
4	0.25	4.8527		

$$I \cong \frac{4}{3} I(h_2) - \frac{1}{3} I(h_1)$$

$$I \cong \frac{16}{15} I_m - \frac{1}{15} I_l$$

Apply the trapezoid rule, using 3 different step sizes (1, 0.5, 0.25)

x	f(x)	trapezoid
1.0	4.0000	
2.0	6.2500	5.1250
	Total area	5.1250

x	f(x)	trapezoid
1.0	4.0000	
1.5	4.6944	2.1736
2.0	6.2500	2.7361
	Total area	4.9097

x	f(x)	trapezoid
1.00	4.0000	
1.25	4.2025	1.0253
1.50	4.6944	1.1121
1.75	5.3890	1.2604
2.00	6.2500	1.4549
	Total area	4.8527

Based on a stopping criterion εs = 0.5% for the approximate error,
we need 3 iterations (I=4.83347014)

* calculating the errors based on the top row

$$\varepsilon_a = \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} 100\%$$

* using the absolute value

ε_t = abs[(true value - estimated value)/true value] * 100%

22.3 Use Romberg integration to evaluate

$$\int_0^2 \frac{e^x \ \sin \ x}{1 + x^2} \, dx$$

to an accuracy of ε_s = 0.5%. Your results should be presented in the form of Fig. 22.3.

	1	**2**	**3**
n	ε_a →	**7.9715%**	**0.0997%**
1	1.34376994	1.97282684	1.94183605
2	1.81556261	1.94377297	
4	1.91172038		

22.4 Obtain an estimate of the integral from Prob. 22.2, but using two-, three-, and four-point Gauss-Legendre formulas. Compute ε_t for each case on the basis of the analytical solution.

Change of variable:

$$x = \frac{2+1}{2} + \frac{2-1}{2} x_d = 1.5 + 0.5x_d$$

$$dx = \frac{2-1}{2} dx_d = 0.5dx_d$$

$$I = \int_{-1}^{1} \left(1.5 + 0.5x_d + \frac{1}{1.5 + 0.5x_d}\right)^2 0.5dx_d$$

Therefore, the transformed function is

$$f(x_d) = 0.5\left(1.5 + 0.5x_d + \frac{1}{1.5 + 0.5x_d}\right)^2$$

Two-point formula:

$$I = f\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) = 2.074414 + 2.755961 = 4.830375$$

$$\varepsilon_t = \left|\frac{4.833333 - 4.830375}{4.833333}\right| \times 100\% = 0.0612\%$$

Three-point formula:

$$\begin{aligned}
I &= 0.5555556f(-0.7745967) + 0.8888889f(0) + 0.5555556f(0.7745967) \\
&= 0.5555556(2.022895) + 0.8888889(2.347222) + 0.5555556(2.921322) \\
&= 1.123831 + 2.08642 + 1.622957 = 4.833207
\end{aligned}$$

$$\varepsilon_t = \left|\frac{4.833333 - 4.833207}{4.833207}\right| \times 100\% = 0.0026\%$$

Four-point formula:

$$\begin{aligned}
I &= 0.3478548f(-0.861136312) + 0.6521452f(-0.339981044) + 0.6521452f(0.339981044) \\
&\quad +0.3478548f(0.861136312) \\
&= 0.3478548(2.009026) + 0.6521452(2.16712) + 0.6521452(2.573718) \\
&\qquad + 0.3478548(2.997699) \\
&= 0.698849 + 1.413277 + 1.678438 + 1.042764 = 4.833328
\end{aligned}$$

$$\varepsilon_t = \left|\frac{4.833333 - 4.833328}{4.833333}\right| \times 100\% = 0.00010\%$$

22.14 There is no closed form solution for the error function,

$$erf(a) = \frac{2}{\sqrt{\pi}} \int_0^a e^{-x^2}\ dx$$

Use the two-point Gauss quadrature approach to estimate erf(1.5). Note that the exact value is 0.966105.

Change of variable:

$$x = \frac{1.5+0}{2} + \frac{1.5-0}{2} x_d = 0.75 + 0.75 x_d$$

$$dx = \frac{1.5-0}{2} dx_d = 0.75 dx_d$$

$$I = \int_{-1}^{1} \frac{1.5}{\sqrt{\pi}} e^{-(0.75+0.75x_d)^2}\ dx_d$$

Therefore, the transformed function is

$$f(x_d) = \frac{1.5}{\sqrt{\pi}} e^{-(0.75+0.75x_d)^2}$$

Two-point formula:

$$I = f\left(\frac{-1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right) = 0.765382 + 0.208792 = 0.974173$$

$$\varepsilon_t = \left|\frac{0.966105 - 0.974173}{0.966105}\right| \times 100\% = 0.835\%$$

23.1 Compute forward and backward difference approximations of $O(h)$ and $O(h^2)$, and central difference approximations of $O(h^2)$ and $O(h^4)$ for the first derivative of $y = \sin x$ at $x = \pi/4$ using a value of $h = \pi/12$. Estimate the true percent relative error, ε_t, for each approximation.

	x	***f*(*x*)**
x_{i-2}	0.261799388	0.258819045
x_{i-1}	0.523598776	0.5
x_i	0.785398163	0.707106781
x_{i+1}	1.047197551	0.866025404
x_{i+2}	1.308996939	0.965925826

true = $-\sin(\pi/4) = -0.70710678$

The results are summarized as

	first-order	**second-order**
Forward	0.607024424	0.719740883
ε_t(%)	14.154	1.787
Backward	0.791089631	0.726012753
ε_t(%)	11.877	2.674
Centered	0.699057028	0.706996958
ε_t(%)	1.138	0.016

23.8 Compute the first-order central difference approximations of $O(h^4)$ for each of the following functions at the specified location and for the specified step size:

(a) $y = x^3 + 3x - 15$	at $x = 0$,	$h = 0.25$
(b) $y = x^2 \cos x$	at $x = 0.5$,	$h = 0.1$
(c) $y = \tan(x/3)$	at $x = 2$,	$h = 0.5$
(d) $y = sin(0.5\sqrt{x})/x$	at $x = 1$,	$h = 0.2$
(e) $y = e^x + x$	at $x = 3$,	$h = 0.2$

Compare your results with the analytical solutions.

(a)

	x	*f*(*x*)
x_{i-2}	-0.5	-16.625
x_{i-1}	-0.25	-15.7656
x_i	0	-15
x_{i+1}	0.25	-14.2344
x_{i+2}	0.5	-13.375

$$f'(x) = \frac{-(-13.375) + 8(-14.2344) - 8(-15.7656) - 16.625}{12(0.25)} = 3$$

Analytical: $f'(x) = 3x^2 + 3 = 3(0) + 3 = 3$

$$f''(x) = \frac{-(-13.375) + 16(-14.2344) - 30(-15) + 16(-15.7656) - (-16.625)}{12(0.25)^2} = 0$$

Analytical: $f''(x) = 6x = 6(0) = 0$

Therefore, for this case with a 3rd degree polynomial, the results are exact.

(b)

	x	*f*(*x*)
x_{i-2}	0.3	0.08598
x_{i-1}	0.4	0.14737
x_i	0.5	0.219396
x_{i+1}	0.6	0.297121
x_{i+2}	0.7	0.374773

$$f'(x) = \frac{-(0.374773) + 8(0.297121) - 8(0.14737) + 0.08598}{12(0.1)} = 0.75768$$

Analytical: $f'(0.4) = 2x\, cos\, x - x^2\, sin\, x = 2(0.5)\, cos(\,0.5) - 0.5^2\, sin(\,0.5) = 0.757726$

$$\varepsilon_t = \left|\frac{0.757726 - 0.75768}{0.757726}\right| \times 100\% = 0.00608\%$$

$$f''(x) = \frac{-(0.374773) + 16(0.297121) - 30(0.219396) + 16(0.14737) - (0.08598)}{12(0.1)^2} = 0.576893$$

Analytical: $f''(0.4) = 2\, cos\, x - x^2\, cos\, x - 4x\, sin\, x = 2\, cos(\,0.5) - 0.5^2\, cos(\,0.5) - 4(0.5)\, sin(\,0.5) =$ 0.576918

$$\varepsilon_t = \left|\frac{0.576918 - 0.576893}{0.576918}\right| \times 100\% = 0.0045\%$$